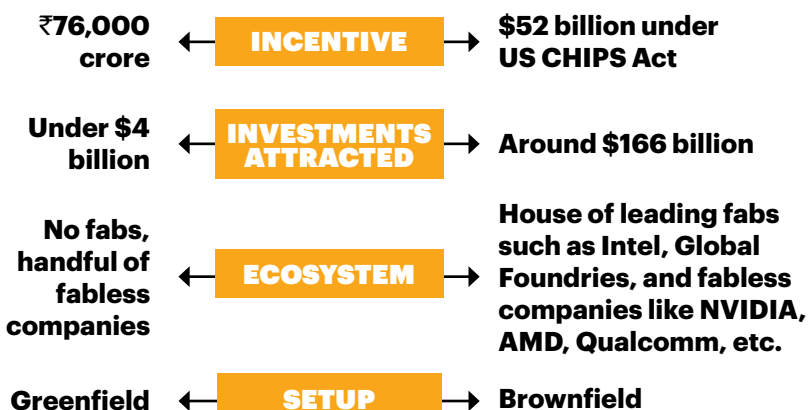


INDIA vs THE US

1 The US has a long history of chip manufacturing and when the US government introduced CHIPS Act, all leading players committed to expansion

2 The US is also close to the customer. Fabs also have local demand from fabless companies like Qualcomm, NVIDIA and electronics companies like Apple, HP, Dell, etc.

3 India is starting from scratch. It has no proven record to attract investments yet



SOURCE PIB, MEITY, IESA, GARTNER, BT RESEARCH

| 91

conductor manufacturing facility here. Unfortunately, the government told him that he could only set up a fab that would not exceed hundreds of thousands of chips, which was a complete non-starter. I have often wondered how the tides with Taiwan and China may have turned had we allowed Intel to set up a fab in India back then!" says Vinod Dham, the father of the Pentium processor. It attempted to do so again with a packaging unit in 2005, but without success. That baggage is troubling now. None of the big chip companies, except for Tower, are considering India.

"India missed the bus for 30-40 years, and an impression was built that we could never do it. India can do it, but is struggling to shrug off the image of having failed in the past

and is unable to get to the 'proven ground' tag," says independent analyst Arun Mampazhy.

Experts and MeitY officials believe that once the ecosystem is set up—which the government is hopeful will happen with Micron's ATMP plant that is being set up at an estimated cost of \$2.7 billion—the floodgates will open.

But ATMP does not involve manufacturing—it is just the testing and packaging part. That comes towards the end of the chain and is preceded by the design and development part and fabrication.

Building those two aspects is easier said than done.

TALENT CRUNCH

One big hurdle is the dearth of skilled workers. It's a problem that

chip companies are grappling with the world over, and even with its engineering base, India lags on this parameter. "Many semiconductor integrated device manufacturer (IDM) /foundry /ATP/ OSAT (outsourced semiconductor assembly and test) players have provided consistent feedback regarding manufacturing and packaging talent unavailability in India as one of the reasons holding back investments," says Danish Faruqi, CEO of Fab Economics, a US-based semiconductor value chain economics consultancy. This puts India's public and private investment in semiconductors at risk of underutilisation.

Quoting a report by the India Brand Equity Foundation, Aditya Joshi—CEO of semiconductor technology, talent, and process so-